Interactformer: Interactive Transformer and CNN for Hyperspectral Image Super-Resolution

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***Abstract*— Due to rich spectral information, hyperspectral images (HSIs) have been widely used in various fields. However, limited by imaging systems, the low spatial resolution of HSIs has become an important problem. In this article, for enhancing the spatial resolution, Interactformer is proposed to interact with global and local features extracted by transformer and 3-D convolutional neural network (CNN) branches. Within the transformer branch, a separable self-attention module with linear complexity is designed to solve the problem that traditional self-attention mechanisms suffer from large memory costs due to quadratic complexity. In the 3-D CNN branch, the spectral attention module and 3-D convolution are applied jointly to better protect the spectral correlation among spectral bands and facilitate local feature extraction of HSIs. The interactive attention unit between the two parallel branches is designed to interact with local and global feature information adaptively. Compared with state-of-the-art super-resolution (SR) methods, the proposed method reconstructs better HSI in simulated SR experiments, real SR experiments, and classification experiments, which proves that Interactformer can effectively improve the spatial resolution while preserving the spectral information.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HSI) super-resolution (SR), interactive attention unit (IAU), transformer.**

I. Introduction

HYPERSPECTRAL images contain hundreds or even thousands of continuous spectral bands covering visible and infrared wavelengths [1], [2]. They provide rich spectral information about the object and the subtle spectral features are reflected. Thus, hyperspectral image (HSI) has become an important and valuable source of information in remote sensing applications such as land cover classification [3], target detection [4], and geological exploration [5]. However, due to the limitation of imaging hardware, the spatial resolution of HSIs is usually low [6]. This restricts the practical application of HSIs in various fields. Hence, it is important to improve the spatial resolution of HSIs.

Super-resolution (SR) [7] solves this problem by increasing the spatial resolution through software algorithms. The process of obtaining a high-resolution (HR) image from a low-resolution (LR) image is SR reconstruction. A multitude of algorithms have been proposed by researchers for HSI SR. They can be categorized as multi-image super-resolution (MISR) [8], [9] and single-image super-resolution (SISR) [10], [11]. The first category (MISR) is fusion-based methods, which generates HR HSI by merging LR HSI and HR auxiliary images such as panchromatic imagery (PAN) [12], [13] or multispectral imagery (MSI) [14], [15]. Dong *et al.* [16] unfolded an iterative SR algorithm into a novel model-guided deep convolutional network for fusing the LR HSIs and HR PANs. The novel structure-color preserving network [17] was designed to deal with the abundant spectral information of LR HSIs and the complex structure characteristics of HR MSIs to obtain SR HSIs. However, these fusion-based methods require auxiliary images (MSI and PAN) with high spatial resolution that may not be readily available. Even with HR auxiliary images, these methods require accurate geometric registration for LR HSIs and auxiliary images. These shortcomings limit the application of fusion-based SR methods.

The second category (SISR) is widely used because it is not limited by the auxiliary image. The main SISR methods include interpolation, low-rank tensor approximation [18], sparse representation [19], and deep learning [20]. Interpolation methods resize or resample an image by simply estimating the unknown pixels based on neighboring pixel values, which is difficult to reconstruct lost high-frequency details and leads to blurred edges. To explore the intrinsic characteristics of HSIs, a novel tensor-based method [18] was presented to achieve spatial SR of HSIs. Huang *et al.* [19] proposed an SR method via multidictionary-based sparse representation to tackle with the HSI restoration problem. However, these methods are inefficient because they are usually formulated as complex optimization problems and must be solved iteratively. Recently, SR methods based on deep learning have acquired favorable performance and are being increasingly studied. The main idea of the SR methods based on deep learning is to learn the nonlinear mapping function between LR images and corresponding HR images. Dong *et al.* [21] first proposed a three-layer convolutional neural network for super-resolution (SRCNN) of natural images. In order to

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obtain better performance, a very deep convolutional network for super-resolution (VDSR) was presented by Kim *et al*. [22]. It shows a significant improvement in accuracy by increasing the depth of the network.

Deep learning-based methods have achieved the state-of-the-art SR performance for natural images due to their powerful learning ability. However, these methods are applicable for natural images with three bands (R, G, and B) and are not directly suitable for HSIs. They usually process each band separately to implement HSI SR, which is difficult to protect the spectral features of HSI data cubes. Hence, some algorithms for HSI SR in particular are desired. Li *et al*. [23] designed a grouped deep recursive residual network (GDRRN) to simultaneously super-resolve all bands of HSIs, which utilized the recursive structure and 2-D convolution. A difference curvature multidimensional network [24] was developed for HSI SR by splitting the input image into several groups to reduce the parameters of the network. Nevertheless, 2-D convolution only focuses on the spatial information of HSIs, and the fixed grouping manner breaks spectral correlation, which leads to spectral disorder. To alleviate the spectral disorder problem, 3-D full convolutional neural network (3D-FCNN) for HSI SR was introduced in [25]. The 3-D convolution can extract spatial–spectral features simultaneously. Hu *et al*. [26] proposed a 3-D multiscale feature fusion network and integrated the spectral loss into training to fully protect spectral correlation. To further improve the texture quality of reconstructed HSIs, two adversarial learning networks [27], [28] were proposed. The adversarial loss is leveraged to guide the generator to produce finer texture details effectively.

However, 3-D networks are often accompanied by a huge number of parameters and high computational complexity. To mitigate the problem in 3-D networks, a mixed 2-D/3-D convolutional network (MCNet) [29] was presented. Compared with using only 3-D convolution, it reduces the parameters and floating point of operations of the network. To reduce the required hardware memory, Wang *et al*. [30] proposed a single-band hyperspectral SR method by aggregating the semantic information from adjacent spectral bands. A spatial–spectral joint SR method [31] was proposed in order to simultaneously enhance the spatial and spectral resolution. These methods focus on exploring local spatial–spectral information. The global features of HSIs are difficult to extract. Accordingly, a hybrid local and nonlocal 3-D attentive CNN (HLNnet) [32] was proposed to extract global features of HSIs first.

Recently, transformer, a model of natural language processing (NLP) [33], has gained increasing popularity in the field of computer vision (CV) such as image classification [34], segmentation [35], and detection [36]. The core of transformer is self-attention, which can explore long-range dependencies. Self-attention mechanism occupies heavy memory, which limits the development of transformer in low-level image processing tasks. Therefore, the self-attention with low computational resources needs to be explored. The GCnet [37] creates a simplified self-attention network based on a query-independent formulation, which restricts the adaptivity of query pixel. Latterly, Liang *et al*. [39] used Swin Transformer [38] for natural image restoration (SwinIR) and achieved satisfactory results. However, it is not suitable for HSI data cubes because the spectral correlation is destroyed. Besides, the CNN-style and transformer-style features are complementary. The combined CNN and transformer network can extract image features competently. STransFuse [40] fuses CNN and Swin Transformer [38] to extract local features while capturing long-range dependencies. Nonetheless, the local and global features are simply fused and lack proper interaction.

To address these drawbacks, an interactive transformer and CNN (Interactformer) is proposed to capture long-range dependencies and extract local feature of HSIs while interacting with each other adaptively. Specifically, a network structure with two parallel branches is proposed by combining transformer and local feature extraction with spectral-preserving module (LFESM). Transformer is composed of multilayer perceptron (MLP), skip connection, and separable self-attention, which effectively extracts global spatial features. It is important to mine global spatial structural information cheaply. The separable self-attention module is designed with linear complexity while better modeling global contexts. LFESM is constructed to extract local features while preserving the spectral correlation. Spectral attention is applied in LFESM for better preserving the spectral signature. Furthermore, the interactive attention unit (IAU) is proposed to consecutively couple the complementary features between the two parallel branches. The interactive manner can substantially refine the local details of global features in the transformer branch and enhance the global perception capability of local features in the 3-D CNN branch. In summary, the main contributions of this work are given as follows.

1) A parallel network called Interactformer is proposed for HSI SR. Transformer branch is used to capture long-range dependencies to obtain global features, and 3-D CNN branch is used to extract local features while preserving the spectral correlation of HSIs.
2) The IAU is designed to interact with local and global features extracted from two parallel branches.
3) The separable self-attention module with linear complexity is designed in the transformer module. The problem of high memory due to quadratic complexity in the traditional self-attention mechanism is solved.
4) The spectral attention module with 3-D convolution in the CNN branch is applied to protect the spectral correlation and extract the spatial–spectral feature effectively.

The rest of this article is divided into three sections. In Section II, the proposed Interactformer is described in detail. Section III is devoted to experiments and results. In Section IV, a conclusion is given.

II. Methodology

In this section, the proposed network is presented in four parts: overall network, IAU, transformer with separable self-attention module, and LFESM.

A. Overall Network

This section gives the overall network structure of the proposed Interactformer. As shown in Fig. 1, the proposed

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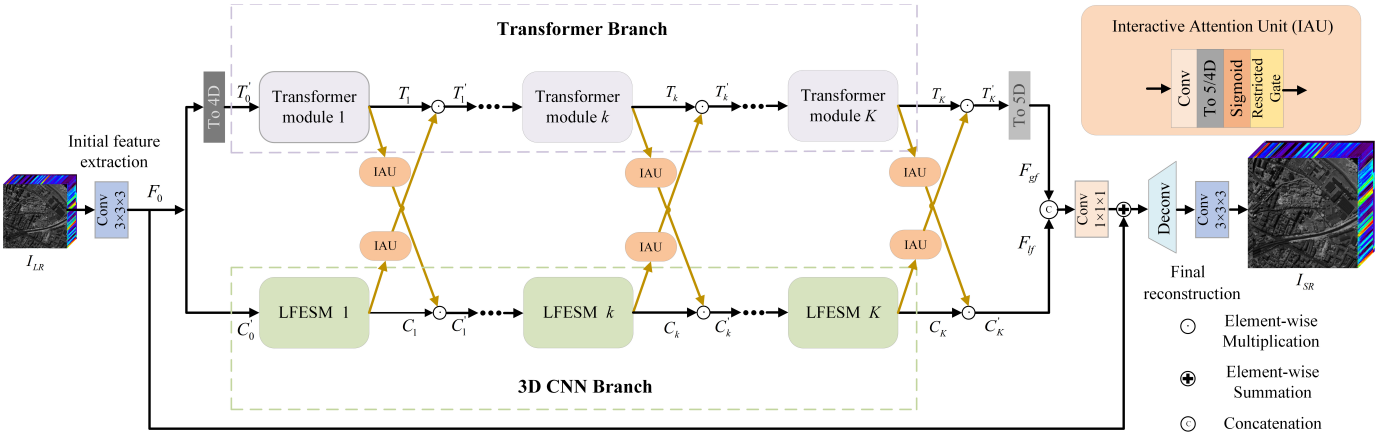


Fig. 1. Overall network structure of the interactive transformer and CNN (Interactformer).

network is mainly composed of two parallel network branches. First, the initial features of input images are extracted by 3-D convolution. Then, the initial feature maps are sent into two parallel branches to extract the global and local features. The long-range dependencies of the feature map are captured by the transformer branch. Local features are extracted and spectral correlation is preserved by LFESMs in the 3-D CNN branch. Furthermore, the IAU is proposed to interact with global and local feature information between two parallel branches. Finally, the reconstruction is performed by deconvolutional upsampling to obtain the SR HSI.

Let $I_{\mathrm{LR}} \in \mathbb{R}^{1\times B\times(H/r)\times(W/r)}$ and $I_{\mathrm{SR}} \in \mathbb{R}^{1\times B\times H\times W}$ denote the input LR HSI and the output SR HSI, respectively, where H and W are the height and width of each band, respectively, B represents the number of bands, r represents the scale factor, and 1 is the number of channels of the feature map. A 3-D convolution with the kernel size of $3 \times 3 \times 3$ is used to initially extract features from I_{LR}

$$F_0 = f_{\mathrm{conv}_{3\times3\times3}}(I_{\mathrm{LR}}). \tag{1}$$

Next, F_0 is reshaped into 4-D data through $f_{\mathrm{reshape4D}}(\cdot)$ and then sent to transformer branch. At the end of the transformer branch, the output is reshaped into 5-D data by $f_{\mathrm{reshape5D}}(\cdot)$. $f_{\mathrm{reshape4D}}(\cdot)$ represents the dimensional transformation from $N \times C \times B \times H \times W$ to $(N \times B) \times C \times H \times W$, C is the number of channels, and N denotes the batch size. $f_{\mathrm{reshape5D}}(\cdot)$ represents the dimensional transformation from $(N \times B) \times C \times H \times W$ to $N \times C \times B \times H \times W$. Meanwhile, the local features are extracted by 3-D CNN branch. Two branches contain K transformer modules and K LFESMs. The output T_k of the kth transformer module and the output C_k of the kth LFESM module can be obtained by

$$\begin{cases} T_k = f_k^T(T'_{k-1}) \\ C_k = f_k^L(C'_{k-1}) \end{cases} \quad (k = 1, \ldots, K) \tag{2}$$

where $f_k^T(\cdot)$ represents the operation of the kth transformer module, $f_k^L(\cdot)$ represents the operation of the kth LFESM, and T'_{k-1} and C'_{k-1} denote the input of the kth transformer module and LFESM, respectively.

Since there is complementary feature between two branches, the global and local features extracted by transformer and 3-D CNN branches are concatenated and expressed as $[F_{gf}, F_{lf}]$, followed by a $1 \times 1 \times 1$ convolution, which can obtain feature maps with long-range dependencies and local information. The final reconstruction can be expressed as follows:

$$I_{\mathrm{SR}} = f_{\mathrm{conv}_{3\times3\times3}}(f_{\mathrm{up}}(f_{\mathrm{conv}_{1\times1\times1}}([F_{gf}, F_{lf}]) + F_0)) \tag{3}$$

where $f_{\mathrm{up}}(\cdot)$ and $f_{\mathrm{conv}_{3\times3\times3}}(\cdot)$ are deconvolutional and convolutional layers.

The loss function is used to measure the reconstruction error. The commonly used loss functions include L_2 loss (mean square error) and L_1 loss (mean absolute error), which measure the pixel difference between SR image and ground truth. Because the L_1 loss does not overpenalize large errors compared with the L_2 loss and shows better convergence performance [41]. We adopt the L_1 loss to measure the reconstruction accuracy. The L_1 loss is given as follows:

$$L_1 = \frac{1}{HWB}\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{b=1}^{B}\|I_{\mathrm{GT}}(i,j,b) - I_{\mathrm{SR}}(i,j,b)\|_1 \tag{4}$$

where I_{GT} and I_{SR} denote the ground truth and the SR image, respectively.

Meanwhile, to protect the spectral correlation of HSIs, the spectral gradient loss [22] is used to further preserve the spectral information. The spectral gradient loss is given as follows:

$$L_G = \frac{1}{HWB}\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{b=1}^{B}\left|I^G_{\mathrm{GT}}(i,j,b) - I^G_{\mathrm{SR}}(i,j,b)\right| \tag{5}$$

where I^G_{GT} and I^G_{SR} denote the spectral gradient of the ground truth and the SR image. The total loss function can be expressed as

$$L_{\mathrm{total}} = L_1 + L_G. \tag{6}$$

B. Interactive Attention Unit

In the previous methods, the interaction of local and global features was usually ignored. In this article, the IAU is

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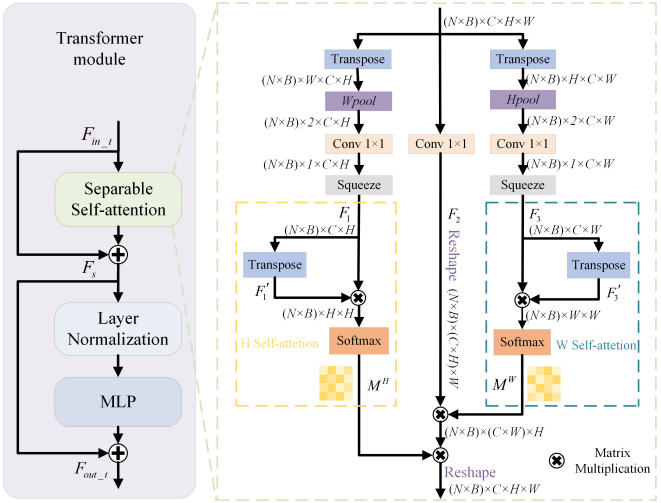


Fig. 2. Internal structure of transformer with separable self-attention module.

designed to interact with local and global features in an adaptive manner.

Specifically, as shown in Fig. 1, IAU can generate weights for global and local features to interact via elementwise multiplication. The size of feature maps in 3-D CNN branch is $N \times C \times B \times H \times W$, and the size of feature maps in transformer branch is $(N \times B) \times C \times H \times W$. Considering the complementarity of the two-style features, the feature information of 3-D CNN branch is transferred to the transformer branch by IAU. First, $1 \times 1 \times 1$ convolution enables cross-channel interaction of feature maps. Then, the feature maps are reshaped to $(N \times B) \times C \times H \times W$ by $f_{\text{reshape4D}}(\cdot)$. To generate weights containing local features and spectral information from 3-D CNN branch, the sigmoid function σ is used to project feature information to weights. Because there exists some difference between local feature and global feature, a simple and straightforward transfer way might cause undesirable results. In order to further enhance the interaction of the two branches, we set the learnable coefficients λ as the restricted gate that can adaptively adjust the weights information from the CNN branch to the transformer branch. Simultaneously, the global spatial information from transformer is transferred to the CNN branch in a similar way. The difference is that the feature maps from the transformer branch require to be reshaped for $N \times C \times B \times H \times W$ by $f_{\text{reshape5D}}(\cdot)$. The specific operation of the IAU between the two branches is represented as

$$\begin{cases} T_k' = \sigma\left(f_{\text{reshape4d}}\left(f_{\text{conv}_{1\times1\times1}}(C_k)\right)\right) \cdot \lambda_k^{\text{CT}} \cdot T_k \\ C_k' = \sigma\left(f_{\text{reshape5d}}\left(f_{\text{conv}_{1\times1}}(T_k)\right)\right) \cdot \lambda_k^{\text{TC}} \cdot C_k \end{cases} \tag{7}$$

where λ_k^{CT} and λ_k^{TC} are the learnable coefficients of the IAU after the kth LFESM and transformer module.

C. Transformer With Separable Self-Attention Module

Fig. 2 shows the proposed transformer module consisting of the separable self-attention module, layer normalization, and the MLP. The recent transformer [42], [43] for CV uses self-attention to capture long-range dependencies among all tokens. Self-attention can directly capture long-distance relationships among pixels to process spatial information in a dynamic manner. We focus on learning the capture process of long-range dependencies in the training phase, and this dynamic manner can help the network to obtain long-range spatial dependencies in the inference phase. Nevertheless, the matrix multiplication of self-attention generates huge attention maps, which contributes to the quadratic computation cost associated with the size of feature maps. It is unsuitable to use the original self-attention module on low-level image processing, which is the main issue for applying transformer to HSI SR. To solve this issue, a separable self-attention module is designed with linear complexity by obtaining global features separably in the H and W dimensions. It is applied for SR to achieve better results.

As we can see, the separable self-attention module is divided into three paths. First, $F_1 \in \mathbb{R}^{C\times H}$ are obtained through sequentially performing the following operation: transpose in the W and C dimensions, pool in the W dimension via $W\text{pool}(\cdot)$, reduce dimension by 1×1 convolution, and squeeze operation. $W\text{pool}(\cdot)$ can be represented as

$$W\text{pool}(\cdot) = [\text{Maxpool}(\cdot), \text{Avgpool}(\cdot)] \tag{8}$$

where the max- and average-pooling operations consist of the $W\text{pool}(\cdot)$. The shape of tensors is changed from $W \times C \times H$ to $2 \times C \times H$ through $W\text{pool}(\cdot)$.

Second, $F_1' \in \mathbb{R}^{H\times C}$ is obtained by transposing $F_1 \in \mathbb{R}^{C\times H}$, which conduct matrix multiplication with F_1. Then, the attention map $M^H \in \mathbb{R}^{H\times H}$ on H dimension is obtained via the softmax operation. This can be expressed specifically as follows:

$$M_{ij}^H = \frac{\exp(F_{1i} \cdot F_{1j})}{\sum_{i=1}^{N\times B} \exp(F_{1i} \cdot F_{1j})} \tag{9}$$

where M_{ij}^H measures the impact of the ith position on the jth position on H dimension. A similar operation is used to obtain the attention map $M^W \in \mathbb{R}^{W\times W}$ to measure the impact of the ith position on the jth position on W dimension. The attention maps M^H and M^W are matrices of comparatively small dimension. Specifically, the computational complexity of the original self-attention is $(HW)^2C$, and the computational complexity of the separable self-attention is $H^2WC + W^2HC$. Due to $H, W \ll HW$, the complexity of the separable self-attention module is linear. Finally, the obtained attention maps are matrix multiplied sequentially with the height and width dimensions of F_2. F_2 is the feature map obtained by a linear operation of 1×1 convolution. The separable self-attention module can obtain spatial information separately along the height and width dimensions. After that, F_s is obtained by skip connection and can be expressed as

$$F_s = f_{\text{SSA}}(F_{\text{in_}t}) + F_{\text{in_}t} \tag{10}$$

where $F_{\text{in_}t}$ represents the input of the transformer module and F_s is the output of the separable self-attention module $f_{\text{SSA}}(\cdot)$. The output $F_{\text{out_}t}$ of the proposed transformer module can be obtained as follows:

$$F_{\text{out_}t} = f_{\text{MLP}}(f_{\text{LN}}(F_s)) + F_s \tag{11}$$

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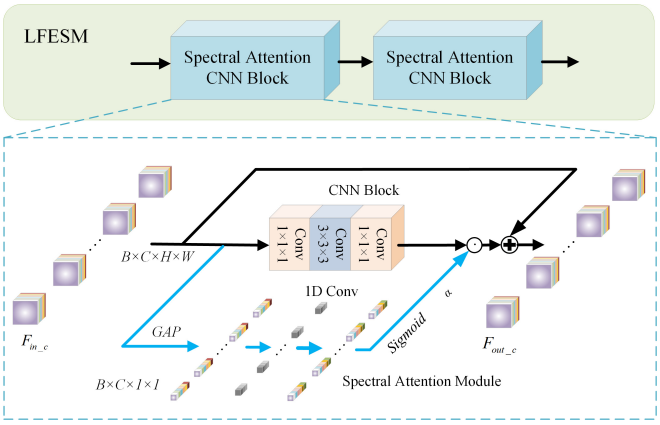


Fig. 3. Internal structure of LFESM and its subblock spectral attention CNN block.

where $f_{\mathrm{LN}}(\cdot)$ represents the layer normalization and $f_{\mathrm{MLP}}(\cdot)$ is a fully connected layer along the channel dimension.

D. Local Feature Extraction With Spectral-Preserving Module

Compared with 2-D networks, 3-D networks can better preserve the spectral features of HSIs. The 3-D CNN branch of the proposed network is used to preserve the spectral feature of HSIs and extract the local feature information concurrently. Aiming to further preserve the spectral features, the spectral attention module is applied to the CNN block.

LFESM contains two spectral attention CNN blocks, which consist of two $1 \times 1 \times 1$ convolutions, one $3 \times 3 \times 3$ convolution, and one spectral attention module, as shown in Fig. 3. The $1 \times 1 \times 1$ convolutions are used for dimension increase and reduction, and the $3 \times 3 \times 3$ convolution is used for spatial–spectral feature extraction. In addition, the global average pooling is used to generate spectral band descriptor by aggregating feature maps across their spatial dimensions ($H \times W$). Next, the 1-D convolution effectuates learning of the spectral relationship to adequately preserve the sophisticated spectral correlation. The bands in different channels of the feature map are regarded as processing different spectral features. Compared with the fully connected structure of the squeeze-and-excitation network (SENet) [44], the learning of spectral relationships by 1-D convolution does not dimensional reduction while attaining better performance and efficiency. Efficient channel attention network (ECANet) [45] achieves channel interaction by 1-D convolution and obtains favorable performance. Therefore, we set the 1-D convolution of size 1×3 to preserve the spectral correlation in the spectral attention module. The sigmoid function is used to generate the spectral attention weight α. Then, α is used to recalibrate the spectral features. The spectral attention CNN block can be formulated as follows:

$$F_{\mathrm{out_}c} = f_{\mathrm{CNN}}(F_{\mathrm{in_}c}) \cdot \alpha + F_{\mathrm{in_}c} \tag{12}$$

where $f_{\mathrm{CNN}}(\cdot)$ is the CNN block and $F_{\mathrm{in_}c}$ and $F_{\mathrm{out_}c}$ represent the input and output of the spectral attention CNN block, respectively.

III. Experiments

In this section, extensive experiments are conducted to evaluate the performance of the proposed Interactformer. We first introduce three public hyperspectral remote sensing datasets. Then, the evaluation metrics and implementation details are described. Next, the ablation experiments are constructed to fully demonstrate the effectiveness of the proposed method. Finally, the advantage of the proposed method is verified by comparisons with state-of-the-art methods.

A. Datasets

1) Pavia Centre Dataset [46]: The Pavia Centre HSI was acquired by the ROSIS sensor during a flight campaign over Pavia, Italy. The number of spectral bands is 102 for Pavia Centre, which is an image of 1096 × 1096 pixels. However, some areas of the image contain no information and have to be discarded. We remove two noisy bands and divide the image into 70% training region (1096 × 515 × 100) and 30% test region (1096 × 200 × 100). The training data are flipped horizontally, rotated (90°, 180°, and 270°), and cut into 7770 samples, which is size of 32 × 32 × 100. The test dataset consists of eight images with the size of 128 × 128 × 100.

2) Chikusei Dataset [47]: The Chikusei hyperspectral dataset has 128 bands in the spectral range from 363 to 1018 nm. The scene consists of 2517 × 2335 pixels and the ground sampling range is 2.5 m. We extract eight nonoverlap regions with 128 × 128 × 128 pixels to form the test data. The remaining area is used as training data, which is cut into 8160 samples of size 32 × 32 × 128.

3) Houston Dataset [48]: The Houston hyperspectral dataset consists of 144 spectral bands in the spectral range from 380 to 1050 nm. Houston dataset is an image of size 349 × 1905 pixels. We divide the image into training region (170 × 1900 × 144) and test region (170 × 1900 × 144). Then, the training region is cut into 7470 samples, and the sample size is 32 × 32 × 144. The test dataset consists of 14 images with the size of 128 × 128 × 144.

The above images are used as the ground truth, while the LR HSIs are simulated by gaussian downsampling with three scale factors, i.e., 2, 4, and 8. Due to the huge memory requirement of HSIs and the hardware limitations, we did not train the network on larger HSIs, and the training set of small images could reduce the training time.

B. Evaluation Metrics and Implementation Details

1) Evaluation Metrics: To qualitatively measure the proposed Interactformer, four evaluation metrics are used to verify the effectiveness of the proposed method in the simulated and real experiments. Specifically, the mean peak signal-to-noise ratio (MPSNR) [49] and mean structure similarity (MSSIM) [50] are universal evaluation metrics for image quality assessment. The former represents the similarity based on MSE between the ground truth and the generated SR result, while the latter describes structural consistency. Their mean values across all spectral bands are used as metrics of spatial

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visual quality. Higher MPSNR and MSSIM values mean better quality. The spectral angle mapper (SAM) [51] evaluates the spectral fidelity, and the optimal value of the SAM is zero. In the real HSIs experiments, the test image is the original HSI without the reference image. Q-metric [52] does not need a reference image and is used to evaluate the SR performance. Higher Q-metric value means better SR performance.

2) Implementation Details: In the proposed Interactformer, the numbers of channels for each transformer module and LFESM are 64. The numbers of transformer module and LFESM are set to 4. In the spectral attention CNN block, the number of channels of feature map is increased to 128 by $1 \times 1 \times 1$ convolution. The kernel size of convolutional layer of the CNN block is $3 \times 3 \times 3$, and the kernel size of convolutional layer of the spectral attention module is 1×3. The weights are initialized by the Kaiming initialization [53]. The network is optimized using ADAM [54] with the default setting. The initial learning rate is set at 0.0005 and halved every 30 epochs. The batch size is set at 16 due to the limitations of GPU memory. The proposed method is performed on NVIDIA GTX 1080Ti GPU using the Pytorch framework.

C. Ablation Study

To verify the effectiveness of the proposed method, a series of ablation experiments is performed on the Pavia Centre datasets with a scale factor of 4. First, the number of modules of parallel branches is set from 2 to 5 to analyze the parameter K. Table I gives the results of the evaluation metrics, which indicates that larger K presents better performance when K is less than 5. Deeper networks allow for better SR reconstruction. However, the evaluation metrics begin to decline when K is set to 5. The main reason for this phenomenon is that the deeper the network is, the harder it is to train. Therefore, the parameter K is set to 4 in our method.

Then, a series of experiments is conducted in order to illustrate the advantages of separable self-attention and spectral attention modules. GCNet [37] is a simplified self-attention network with less computation. The cross-window self-attention (CSA) of Swin Transformer [39] uses local self-attention and cross-window interaction to reduce computational complexity. SENet [44] models the dependencies among channels and recalibrates channelwise features. The global context block (GCB) of GCnet and CSA is used to replace the separable self-attention module to verify global feature extraction capabilities in Interactformer. The squeeze-and-excitation block (SEB) of SENet is used to replace the spectral attention module to compare spectral preservation performance. The experimental results are given in Table II. From the values of the three evaluation metrics, we can observe that the performance of the separable self-attention module is better than that of GCB. Meanwhile, the separable self-attention is superior to CSA, which restricts context aggregation of local neighborhoods. Compared with SEB, the proposed spectral attention module can significantly reduce the SAM value of the reconstructed image and achieve favorable results. Meanwhile, the evaluation metrics of single transformer branch, single CNN branch, parallel transformer branches, and parallel CNN branches are also given in Table II. We can observe that their evaluation metrics are poorer than the Interactformer. The single transformer branch lacks local feature extraction and protection of spectral features, and the single CNN branch lacks extraction of global features, which causes performance degradation. The parallel transformer branches and parallel CNN branches present worse evaluation metric values than the Interactformer because the global and local features are not extracted simultaneously.

TABLE I
STUDY ON THE NUMBER OF MODULES OF PARALLEL NETWORKS ON THE PAVIA CENTRE DATASET (SCALE FACTOR 4)

	2	3	4	5
MPSNR	28.38	28.41	28.51	28.48
MSSIM	0.8868	0.8856	0.8897	0.8890
SAM	5.96	5.70	5.61	5.72

TABLE II
NETWORK STRUCTURE ANALYSIS ON THE PAVIA CENTRE DATASET (SCALE FACTOR 4)

	MPSNR	MSSIM	SAM
Separable self-attention module is replaced with GCB	28.44	0.8875	5.71
Separable self-attention module is replaced with CSA	28.41	0.8870	5.70
Spectral attention module is replaced with SEB	28.42	0.8872	5.74
Single Transformer branch	28.07	0.8763	5.76
Single CNN branch	28.35	0.8862	5.76
Parallel Transformer branches	28.29	0.8832	5.81
Parallel CNN branches	28.45	0.8875	5.75
Interactformer	28.51	0.8897	5.61

TABLE III
ABLATION STUDY WITH IAU AND SPECTRAL ATTENTION MODULE ON THE PAVIA CENTRE DATASET (SCALE FACTOR 4)

Interactive attention unit	Spectral attention module	MPSNR	MSSIM	SAM
×	×	28.34	0.8857	5.98
×	√	28.36	0.8856	5.75
√	×	28.47	0.8894	5.88
√	√	28.51	0.8897	5.61

To further validate the ability of the feature interaction and spectral protection in the proposed method, we conducted ablation experiments on IAUs and spectral attention. The experimental results are given in Table III. The results show that MPSNR significantly decreases after removing the IAU. The removal of the spectral attention module in the CNN branch leads to an increase of the SAM values, which can be derived as a decrease in the spectral protection capability. The evaluation metric values become worse and the network performance decreases when the IAU and spectral attention module are removed simultaneously. Therefore, these components can effectively improve the SR performance.

D. Simulated HSI SR Experiments

In this section, eight methods are selected as the baseline for comparison: Bicubic, VDSR [22], GDRRN [23],

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TABLE IV

QUANTITATIVE COMPARISON OF THE PAVIA CENTRE DATASET. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST VALUE, RESPECTIVELY

Scale	Metric	Bicubic	GDRRN	3D-FCNN	VDSR	SwinIR	HLNnet	SSPSR	MCNet	Ours
×2	MPSNR	30.95	32.40	33.75	33.42	33.54	34.57	-	34.55	34.78
	MSSIM	0.9337	0.9460	0.9661	0.9591	0.9595	0.9719	-	0.9718	0.9729
	SAM	4.62	5.34	4.01	4.47	4.23	3.80	-	3.81	3.69
×4	MPSNR	26.80	27.41	28.07	27.37	27.92	28.20	28.28	28.32	28.51
	MSSIM	0.7851	0.8354	0.8763	0.8411	0.8553	0.8832	0.8859	0.8860	0.8897
	SAM	6.26	7.23	5.75	7.56	6.09	6.18	5.63	5.95	5.61
×8	MPSNR	24.20	24.56	24.70	24.40	24.65	24.54	24.69	24.72	24.83
	MSSIM	0.6123	0.6388	0.6880	0.6386	0.6509	0.6836	0.6956	0.6990	0.7195
	SAM	7.73	8.13	7.71	8.83	7.74	8.24	7.89	7.98	7.70

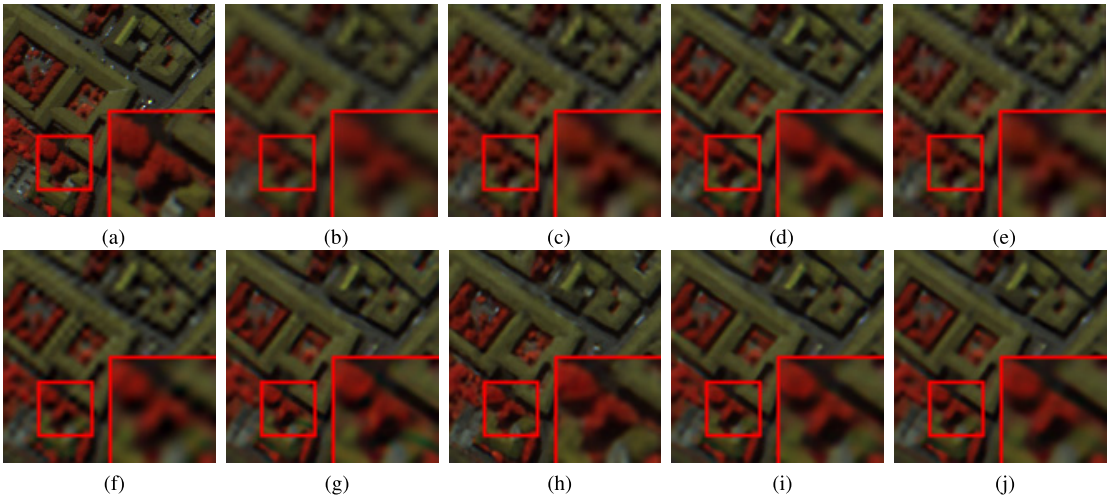


Fig. 4. HSI SR results in the Pavia Centre dataset (scale factor 4). (a) Ground truth. (b) Bicubic. (c) GDRRN. (d) 3D-FCNN. (e) VDSR. (f) SwinIR. (g) HLNnet. (h) SSPSR. (i) MCNet. (j) Ours. The false-color image is used for clear visualization (red: 100, green: 30, and blue: 10).

3D-FCNN [25], SwinIR [39], HLNnet [32], learning spatial–spectral prior for HSI SR (SSPSR) [10], and MCNet [29]. Bicubic is a classic interpolation method, VDSR is the SR method based on CNNs, and SwinIR is the state-of-the-art SR methods based on transformer. These methods are representative CNN- and transformer-based for natural image SR. In order to apply the SR method of natural images to HSIs, a band-by-band method is used. Specifically, the HSI is considered as a series of independent gray images. 3D-FCNN, GDRRN, HLNnet, SSPSR, and MCNet are deep learning methods for HSI SR. The simulated experiments are conducted on three datasets with three scale factors, i.e., 2, 4, and 8. Due to the progressive upsampling network of SSPSR, we only conducted experiments with scale factors of 4 and 8 for SSPSR.

1) Pavia Centre Dataset: Table IV tabulates the quantitative results of different SR methods for different scale factors on the Pavia Centre dataset. The red and blue represent the best and the second-best results, respectively. It is observed that the proposed Interactformer outperforms all other SR methods. When the scale factor is 2, the MPSNR of our method is 0.23 dB higher than that of MCNet, and the result of SAM is 0.11 lower than that of HLNnet. Among all the competing methods GDRRN, VDSR, SwinIR, and SSPSR are based on 2-D networks, while other methods are based on 3-D networks for HSI SR. Since the 3-D network can jointly extract spatial–spectral features, the image reconstruction performance is better than that of the 2-D network. Interactformer can jointly extract not only spatial–spectral features but also global and local features interactively to obtain better SR results.

The above quantitative analysis is also supported by the qualitative results shown in Figs. 4 and 5. In terms of image reconstruction quality, the proposed method produces sharper edges compared with other compared methods. Fig. 4 enlarges an area marked by the red rectangle to show more details. We can observe that the proposed method reconstructs more accurate results such as the edge details, while other methods restore unclear boundaries (MCNet) and appear spectral distortion (HLNnet) and significant artifacts (other comparison methods). In order to further illustrate the advantages of our method, we visualize the error images to intuitively display the accuracy of the reconstruction for each pixel in Fig. 5. The smaller the error is, the closer the reconstructed image will be to the ground truth. The bluer the absolute error map is, the better the reconstructed image will be. The proposed method achieves the best reconstruction performance and minimum error.

2) Chikusei Dataset: The quantitative and qualitative results of Chikusei dataset are shown in Table V and Figs. 6 and 7,

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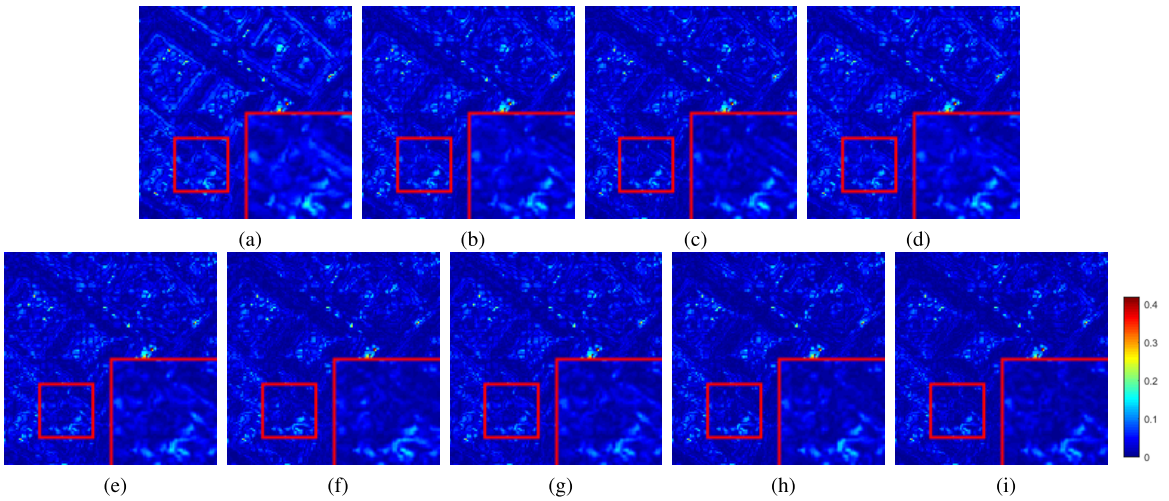


Fig. 5. Error images between the reconstructed images and the ground truth in the Pavia Centre dataset (scale factor 4). (a) Bicubic. (b) GDRRN. (c) 3D-FCNN. (d) VDSR. (e) SwinIR. (f) HLNnet. (g) SSPSR. (h) MCNet. (i) Ours.

TABLE V

QUANTITATIVE COMPARISON OF THE CHIKUSEI DATASET. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST VALUE, RESPECTIVELY

Scale	Metric	Bicubic	GDRRN	3D-FCNN	VDSR	SwinIR	HLNnet	SSPSR	MCNet	Ours
×2	MPSNR	31.98	34.77	34.99	35.20	35.88	36.21	-	36.29	36.37
	MSSIM	0.9676	0.9850	0.9909	9.9872	0.9916	0.9929	-	0.9930	0.9931
	SAM	1.98	1.60	1.55	1.63	1.42	1.36	-	1.35	1.33
×4	MPSNR	26.80	27.53	28.13	28.39	28.62	28.93	28.95	28.92	29.06
	MSSIM	0.8940	0.9148	0.9569	0.9305	0.9335	0.9630	0.9632	0.9625	0.9635
	SAM	3.63	3.54	3.10	3.06	3.13	2.89	2.85	2.92	2.77
×8	MPSNR	23.72	24.20	24.25	24.15	24.20	24.28	24.30	24.34	24.56
	MSSIM	0.7741	0.8130	0.8907	0.8127	0.8178	0.8916	0.8918	0.8920	0.8974
	SAM	5.49	5.19	5.04	5.24	5.38	5.14	5.02	5.09	4.81

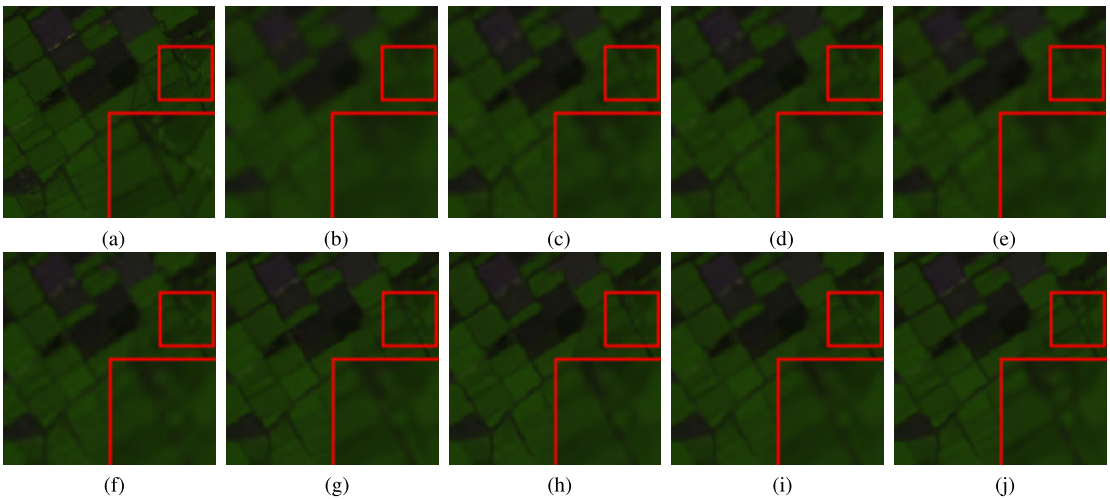


Fig. 6. HSI SR results in the Chikusei dataset (scale factor 4). (a) Ground truth. (b) Bicubic. (c) GDRRN. (d) 3D-FCNN. (e) VDSR. (f) SwinIR. (g) HLNnet. (h) SSPSR. (i) MCNet. (j) Ours. The false-color image is used for clear visualization (red: 70, green: 100, and blue: 36).

respectively. As shown in Table V, the proposed method is superior to other methods in three evaluation metrics with two, four, and eight scale factors. We can find that as the scale factor increases, the proposed method has an increasing advantage over other methods. The main reason may be that the proposed method interacts with local and global features adaptively in the reconstruction process, which can better mine feature information of degraded images with large-scale factor. Also, the SAM value of our method is the lowest in all scale factors. The reason is that the 3-D CNN branch can better

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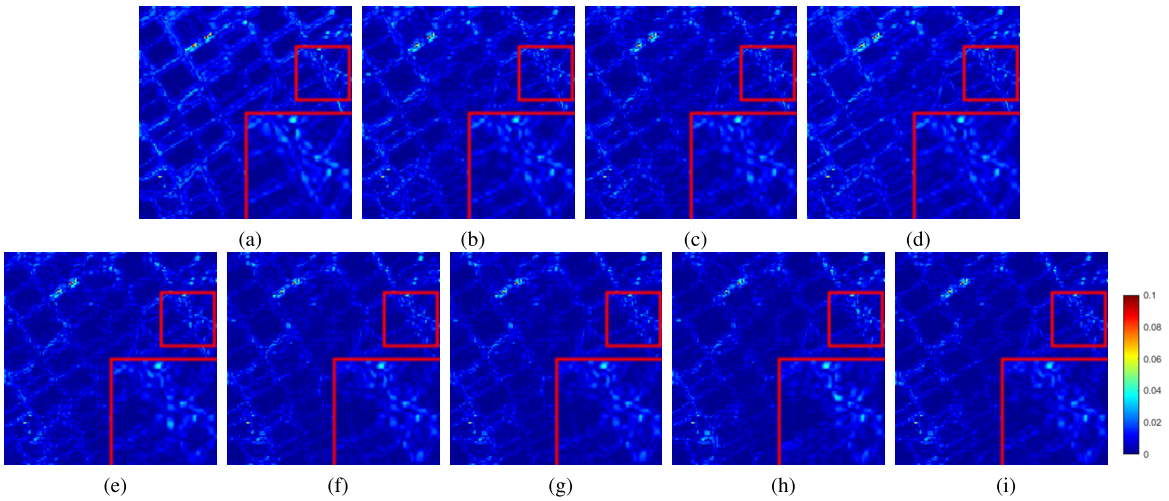


Fig. 7. Error images between the reconstructed images and the ground truth in the Chikusei dataset (scale factor 4). (a) Bicubic. (b) GDRRN. (c) 3D-FCNN. (d) VDSR. (e) SwinIR. (f) HLNnet. (g) SSPSR. (h) MCNet. (i) Ours.

TABLE VI

QUANTITATIVE COMPARISON OF THE HOUSTON DATASET. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST VALUE, RESPECTIVELY

Scale	Metric	Bicubic	GDRRN	3D-FCNN	VDSR	SwinIR	HLNnet	SSPSR	MCNet	Ours
×2	MPSNR	31.59	32.92	32.94	33.04	32.66	33.26	-	33.53	33.71
	MSSIM	0.9457	0.9458	0.9628	0.8463	0.9429	0.9642	-	0.9665	0.9681
	SAM	3.59	3.28	3.01	3.16	3.36	3.09	-	2.95	2.79
×4	MPSNR	28.66	28.99	29.36	29.18	29.21	29.64	29.58	29.52	29.74
	MSSIM	0.8299	0.8403	0.9122	0.8629	0.8632	0.9166	0.9160	0.9141	0.9181
	SAM	5.87	6.08	5.09	5.57	5.14	4.98	5.02	5.07	4.75
×8	MPSNR	26.60	26.79	26.81	26.85	26.63	26.87	27.00	27.05	27.09
	MSSIM	0.7046	0.7433	0.8200	0.7433	0.7574	0.8402	0.8438	0.8444	0.8429
	SAM	8.24	8.06	7.75	7.90	7.96	7.86	7.57	7.66	7.48

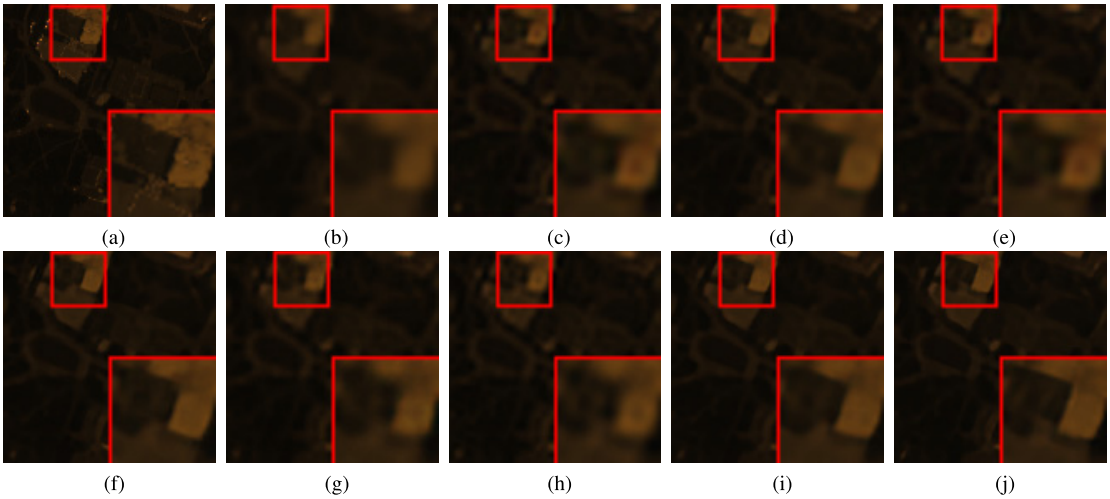


Fig. 8. HSI SR results in the Houston dataset (scale factor 4). (a) Ground truth. (b) Bicubic. (c) GDRRN. (d) 3D-FCNN. (e) VDSR. (f) SwinIR. (g) HLNnet. (h) SSPSR. (i) MCNet. (j) Ours. The false-color image is used for clear visualization (red: 22, green: 14, and blue: 8).

preserve the spectral signature through 3-D convolution and spectral attention module.

To evaluate the perceptual quality of the reconstructed image, Fig. 6 shows the reconstructed images from the Chikusei dataset and enlarges the area marked by the red rectangle to show in detail. The proposed method generates more realistic visual effect and sharper edges, while other competing methods fail to recover these details. Fig. 7 shows the error images for all methods on the Chikusei dataset. It can be observed that the reconstructed result of our method

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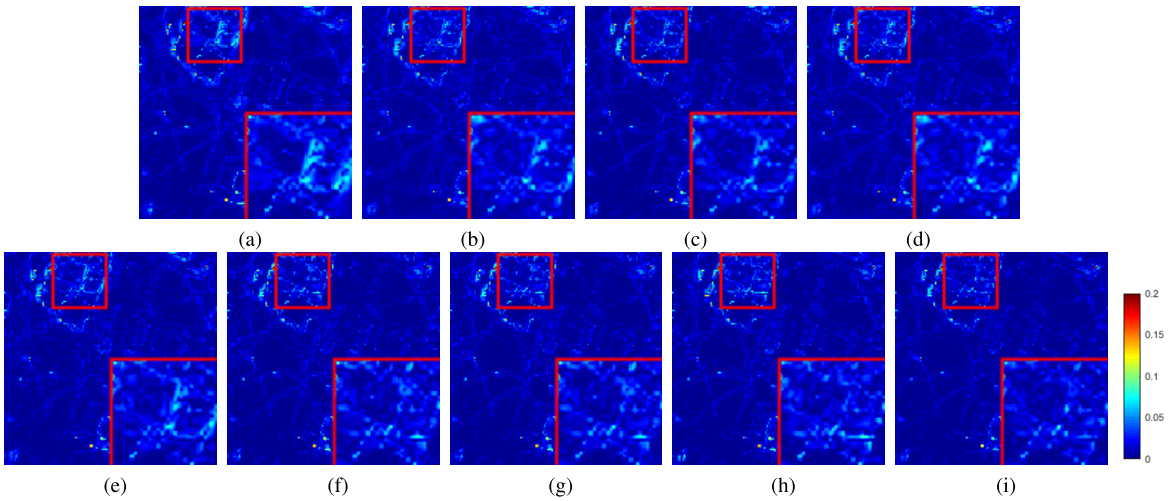


Fig. 9. Error images between the reconstructed images and the ground truth in the Houston dataset (scale factor 4). (a) Bicubic. (b) GDRRN. (c) 3D-FCNN. (d) VDSR. (e) SwinIR. (f) HLNnet. (g) SSPSR. (h) MCNet. (i) Ours.

presents less error. Thus, from the perspective of reconstructed image quality, our method achieves the best.

3) Houston Dataset: We further conduct experiments on the Houston dataset to verify the performance of Interactformer in Table VI. The Houston dataset contains more spectral bands, so preserving the image spectral signature is more difficult. Nevertheless, 2-D networks (i.e., GDRRN, VDSR, SwinIR, and SSPSR) usually perform poorly in protecting spectral features. Therefore, these methods obtain larger SAM values. The proposed method achieves not only best SAM criteria but also lead to better performance in both MPSNR and MSSIM values, where the MPSNR value is 0.22 higher than MCNet with the scale factor of 4. This is because the proposed method preserves spectral correlation and extracts better feature, resulting in the best reconstruction performance.

Fig. 8 shows the visual exhibition of the reconstructed HSIs. The results of 2-D CNN (i.e., VDSR and GDRRN) are similar, all of which are blurry. SwinIR can recover some details attributed to the global features that can be effectively extracted through the shifted window but still produce some blurred edges. MCNet reconstructs some edge details, but this is not enough. The proposed method can restore clearer edges. In Fig. 9, we visualize the error image in the same areas to compare the reconstructed performance among SR images in an intuitive way. We can observe that the proposed method obtains lower error. Thus, our method presents a better SR image for the Houston dataset.

Furthermore, the spectral curves can reflect the values of different spectral bands, and the different curves can show the absolute value of the pixel difference between reconstructed images and ground truth, as shown in Fig. 10. The pixel is selected randomly from the test image to compare the spectral similarity between ground-truth and SR images in three datasets. The difference curves are the spectral error between the results of various methods and ground truth. Our method is the orange dashed line. We can observe that Interactformer obtains spectral curves closer to the corresponding ground truth and the difference curves closer to zero, which indicates that the spectral signature is preserved well.

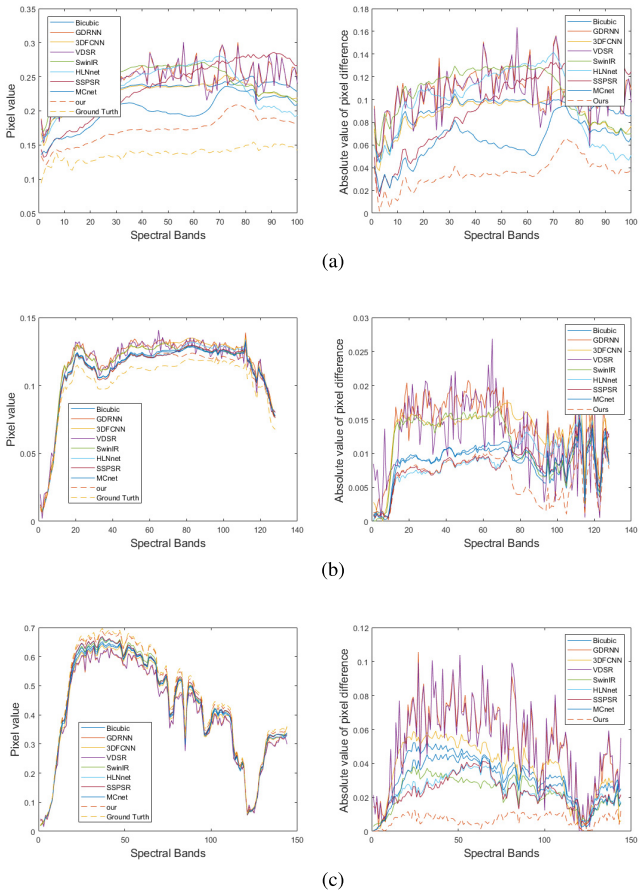


Fig. 10. Example spectral curves and difference curves on a selected pixel value of three datasets with the scale factor of 4. (a) Pavia Center. (b) Chikusei. (c) Houston.

The spectral curves of other methods (i.e., MCNet, HLNnet, SwinIR, and SSPSR) are inconsistent with the ground truth and show spectral distortion.

E. Real HSI SR Experiments

In this section, the Interactformer is applied to real HSIs to identify the performance. The test images are used without degradation operation. For the datasets of Pavia Centre,

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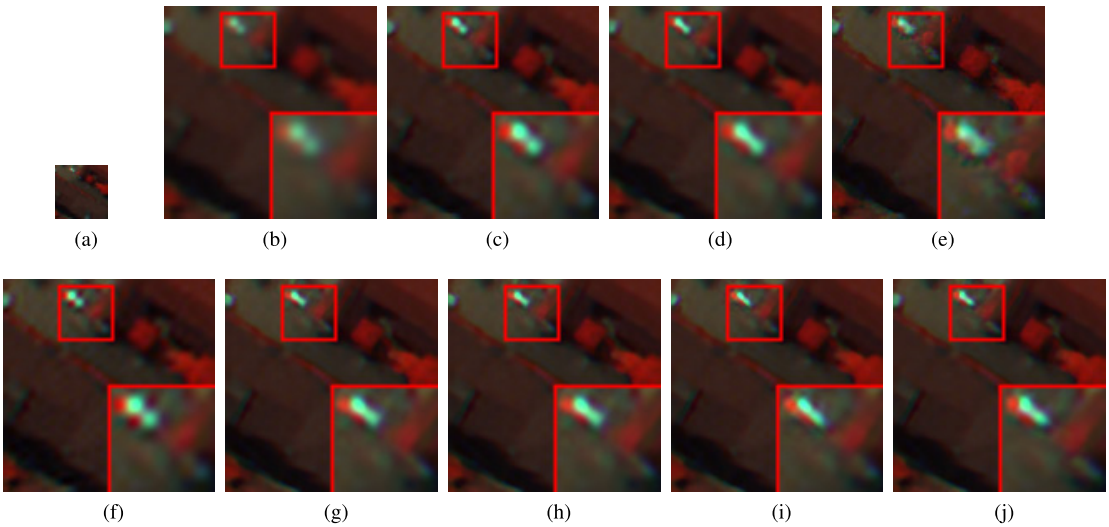


Fig. 11. Real HSI SR results in the Pavia Centre dataset (scale factor 4). (a) Original HSI. (b) Bicubic. (c) GDRRN. (d) 3D-FCNN. (e) VDSR. (f) SwinIR. (g) HLNnet. (h) SSPSR. (i) MCNet. (j) Ours. The false-color image is used for clear visualization (red: 100, green: 30, and blue: 10).

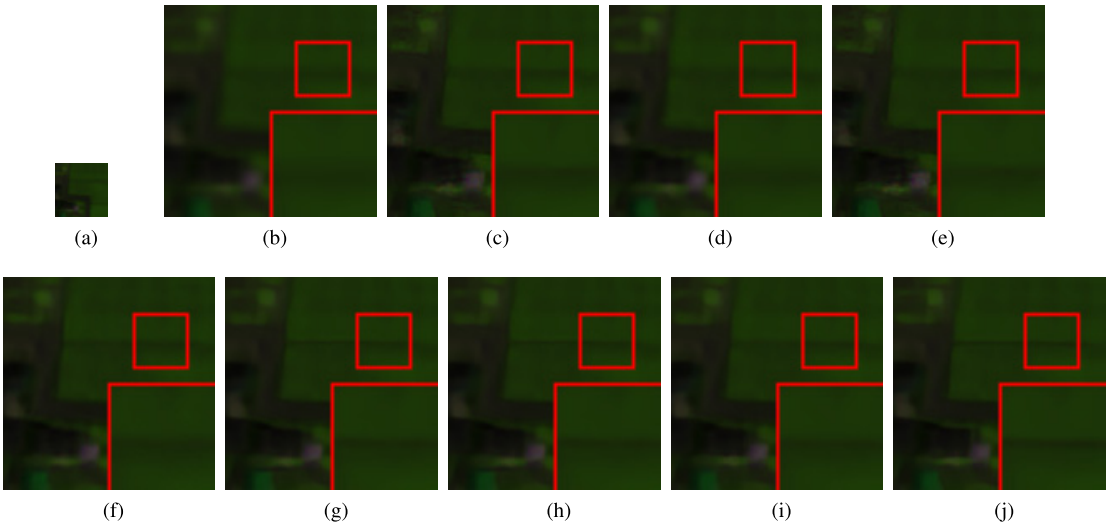


Fig. 12. Real HSI SR results in the Chikusei dataset (scale factor 4). (a) Original HSI. (b) Bicubic. (c) GDRRN. (d) 3D-FCNN. (e) VDSR. (f) SwinIR. (g) HLNnet. (h) SSPSR. (i) MCNet. (j) Ours. The false-color image is used for clear visualization (red: 70, green: 100, and blue: 36).

Chikusei, and Houston, 12 areas with a height and width of 32 × 32 pixels are selected. All methods are applied to these regions with a scale factor of 4 in Figs. 11–13. The network trained in the simulated experiments is employed in the real HSI SR experiments. From the images of the real SR experiment, the proposed method reconstructs clearer points of light in Fig. 11. The results of MCNet and HLNnet are similar, both of which are blurry. In contrast, our method can recover sharper edges in Fig. 12. Also, the proposed method generates more realistic visual quality, such as the clearer contours in Fig. 13.

To further confirm the advantages of the proposed method on real image SR of HSIs, the quality assessment of the real HSI SR images for all methods is shown in Table VII. It can be seen that the best values of Q-metric [52] are obtained by our method for all datasets, which is consistent with the results of the simulation experiments. Specifically, compared with the second-best method MCNet, our method is 0.23 higher in Q-metric on the Pavia Centre dataset and 0.07 higher in Q-metric on the Chikusei dataset, and the results are consistent with the performance of Figs. 11 and 12. On the Houston dataset, our Q-metric is 0.14 higher than the second-best method SSPSR, which is also coincident with the results in Fig. 13. This promotely validates that Interactformer achieves superior SR results.

F. Classification Experiments

In order to better verify the advantages of the Interactformer, we conducted classification experiments on the Pavia Centre dataset. For classification experiments, we need to obtain SR images of different methods. The original HSI is cut into a size of 1096 × 714 × 100 and degenerated to 548 × 357 × 100. Then, the degraded image is put into the model trained by different methods for SR. Each method will get a reconstructed HSI with size 1096 × 714 × 100.

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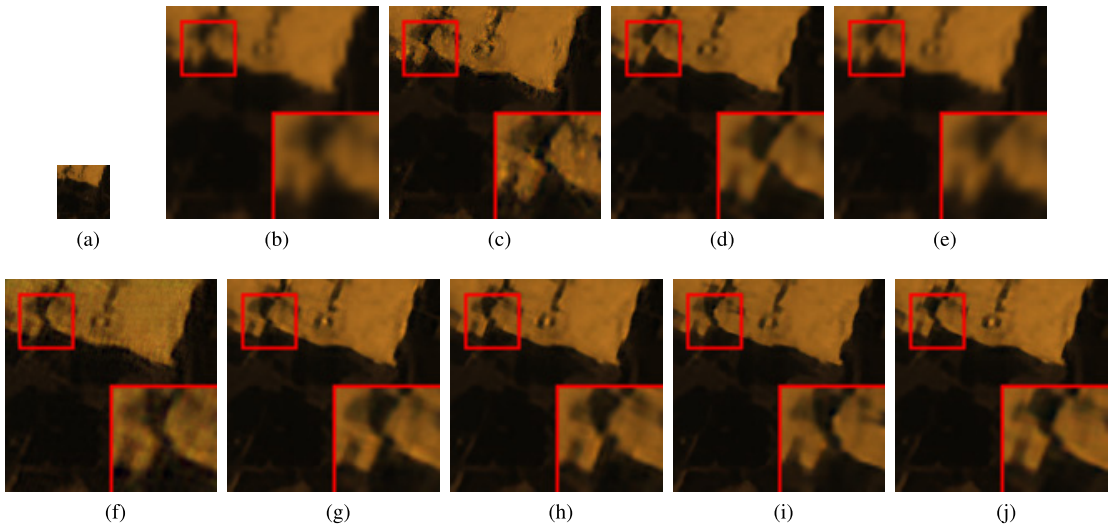


Fig. 13. Real HSI SR results in the Houston dataset (scale factor 4). (a) Original HSI. (b) Bicubic. (c) GDRRN. (d) 3D-FCNN. (e) VDSR. (f) SwinIR. (g) HLNnet. (h) SSPSR. (i) MCNet. (j) Ours. The false-color image is used for clear visualization (red: 22, green: 14, and blue: 8).

TABLE VII

REAL HSI SR EXPERIMENTS IN THREE DATASETS (SCALE FACTOR 4). THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST VALUE, RESPECTIVELY

Dataset	Bicubic	GDRRN	3D-FCNN	VDSR	SwinIR	HLNnet	SSPSR	MCNet	Ours
Pavia Centre	6.70	9.54	9.83	9.61	9.77	10.47	10.21	10.60	10.83
Chikusei	3.46	3.97	4.10	4.49	4.63	4.82	5.05	5.12	5.19
Houston	4.20	6.08	5.93	5.83	6.76	8.45	8.50	8.33	8.64

TABLE VIII

CLASSIFICATION RESULTS OF SVM OVER SR IMAGES OF DIFFERENT METHODS ON PAVIA CENTRE DATASET. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST VALUE IN ADDITION TO THE GROUND TRUTH, RESPECTIVELY

Class NO.	Class Name	Train	Test	Ground -truth	Bicubic	GDRRN	3D-FCNN	VDSR	SwinIR	HLNnet	MCNet	Ours
1	Water	659	65831	1	1	0.96	0.98	1	0.99	0.99	1	0.99
2	Trees	76	7598	0.95	0.99	0.93	0.73	0.77	0.75	0.82	0.84	0.93
3	Meadows	30	3041	0.86	0.38	0.80	0.62	0.96	0.92	0.82	0.89	0.90
4	Bricks	27	2685	0.71	0.29	0.41	0.36	0.40	0.44	0.51	0.34	0.42
5	Bare soil	66	6582	0.93	0.72	0.86	0.51	0.68	0.83	0.88	0.88	0.92
6	Asphalt	92	9242	0.94	0.93	0.51	0.19	0.65	0.58	0.55	0.73	0.78
7	Bitumen	73	7287	0.92	0.64	0.59	0.40	0.94	0.92	0.81	0.55	0.64
8	Tiles	428	42826	0.99	0.99	1	0.96	1	0.99	0.99	0.99	0.99
9	Shadows	29	2863	1	1	1	1	1	1	1	1	1
	AA			0.93	0.77	0.78	0.64	0.82	0.83	0.82	0.80	0.84
	OA			0.98	0.84	0.91	0.85	0.94	0.93	0.92	0.93	0.95
	Kappa			0.97	0.82	0.85	0.72	0.85	0.85	0.84	0.86	0.90

Then, support vector machine (SVM) with an RBF kernel is used as a classification method. The training samples of SVM come from the original HSIs, and we randomly selected 1% of the samples as training data. In order to prevent randomness, we trained and tested ten times, and the final indexes are averaged. Table VIII lists the classification accuracy of different methods in terms of per-class accuracy, average accuracy (AA), overall accuracy (OA), and Kappa coefficient. It can be seen that the classification results of SR HSI reconstructed by our method are significantly better than other methods. The classification map of these reconstructed images and the ground-truth classification map are shown in Fig. 14. It can be clearly seen that the classification map of the proposed method is closer to the classification map of ground truth.

G. Computational Cost

To analyze the computational cost of all the compared algorithms, the number of parameters (Params) and the inference time for all methods are given in Table IX. Specifically, we obtain the Params and inference time for all methods in the Pavia Centre dataset with the scale factor 4. All methods are implemented by Pytorch and the computational cost is measured on a GeForce GTX 1080Ti GPU. As can

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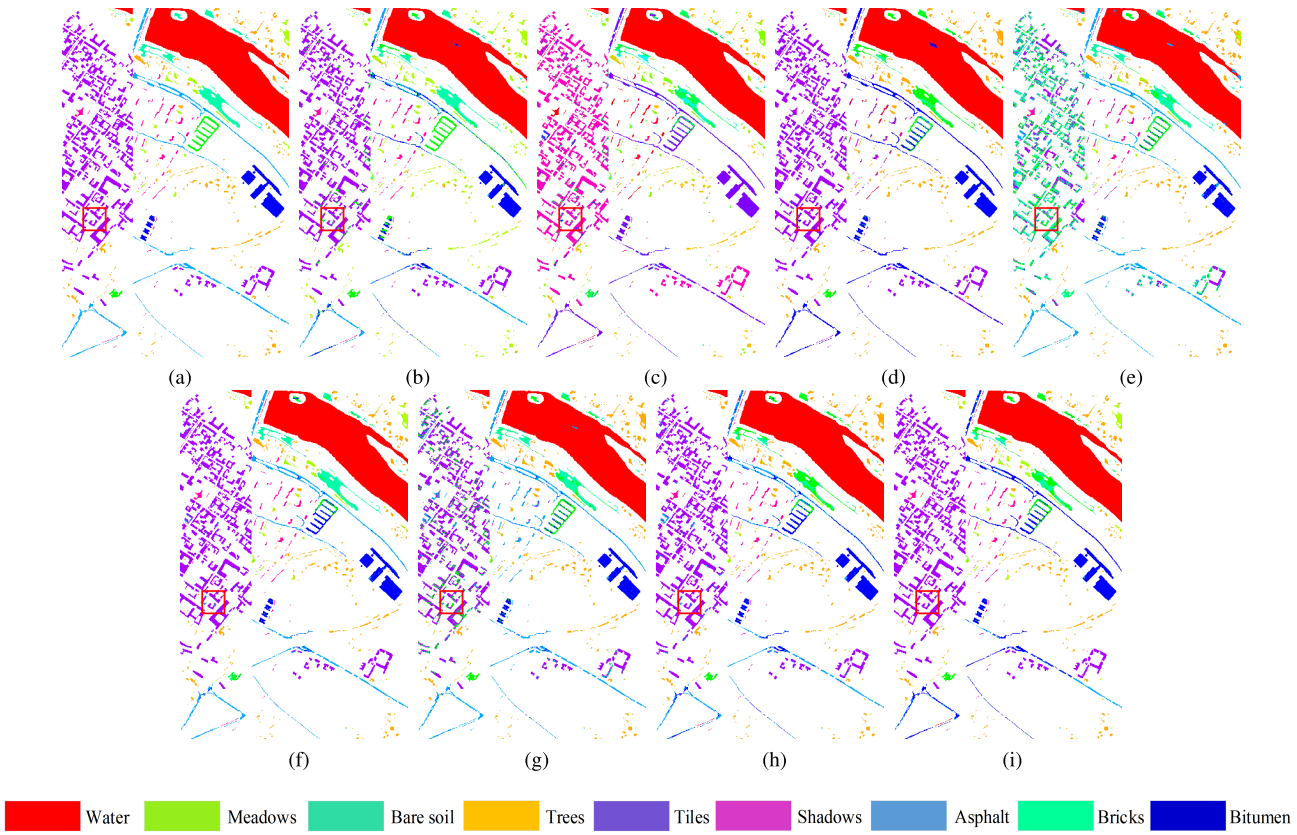


Fig. 14. Classification maps obtained by SVM. (a) Ground truth. (b) Bicubic. (c) GDRRN. (d) 3D-FCNN. (e) VDSR. (f) SwinIR. (g) HLNnet. (h) MCNet. (i) Ours.

TABLE IX
NUMBER OF PARAMETERS AND THE INFERENCE TIME FOR ALL METHODS ON THE PAVIA CENTRE DATASET (SCALE FACTOR 4)

	GDRRN	3D-FCNN	VDSR	SwinIR	HLNnet	SSPSR	MCNet	Ours
Params(K)	219	39	665	11896	9899	12875	2174	4465
Inference Time(s)	0.87	0.90	1.22	0.70	0.63	1.10	0.61	0.67

be seen from Table IX, the inference time is longer for the pre-upsampling methods (i.e., GDRRN, 3D-FCNN, and VDSR). The 2-D networks require more channels to obtain good performance, resulting in a large number of parameters (i.e., SwinIR and SSPSR). The proposed method can obtain better performance with a slightly increased computational cost.

IV. CONCLUSION

In this article, Interactformer is proposed for HSI SR to interact with local and global features. Within the Interactformer, we leverage the transformer module to capture global features and the LFESM to extract local features. In the transformer module, the separable self-attention module is designed to solve the problem of quadratic complexity in the traditional self-attention mechanism. In the LFESM, 3-D convolution and spectral attention module can extract local features while better preserving the spectral signature. IAU between parallel networks is designed to fuse local and global features, enhancing the complementarity of two parallel branches in an interactive manner. Experimental results on three HSI remote sensing datasets show that the proposed Interactformer achieves the best values for different evaluation metrics and obtains better reconstructed images compared with other SR methods. Moreover, the SR result of our method obtains a higher classification accuracy than other methods, which shows the potential for practical applications. In future work, we will explore the extensions of the proposed method to other image restoration applications such as HSI denoising.

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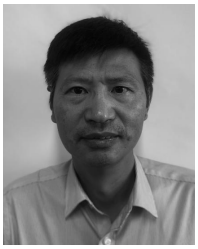
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